

Haokai Ding

UNDERGRADUATE RESEARCHER · ROBOTICS

Shenzhen, China

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Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.Sc. IN ROBOTICS (INCOMING)

Abu Dhabi, UAE

Fall 2026

Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

Shenzhen, China

Sep. 2023 – Present

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Shenzhen, China

Sep. 2022 – Jul. 2026

- GPA: 84.6/100. Research focus: LIBS spectroscopy and robotics systems.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Shanghai, China

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

2023 – Present

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen Technology University

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY-METAL ANALYSIS

Shenzhen, China

2022 – 2023

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

Publications

† Co-first author * Corresponding author Listed in reverse chronological order by IEEE Xplore date added.

SELECTED CONFERENCE PAPERS (ROBOTICS)

Y. Chen, H. Luan, H. Ding, W. Zhang*

IEEE RAAI 2025, Singapore

[P1] AN IDLE-STROKE UNDERACTUATED GRIPPER WITH INDEPENDENT DOUBLE-PARALLELOGRAM LINKAGES FOR PINCHING AND ADAPTIVE GRASPING

Mar. 2026

H. Ding†, X. Zhong, W. Huang, T. Yang*, W. Zhang*

IEEE ROBIO 2025, Bangkok, Thailand

[P2] BIOCREST – A FLEXIBLE ADAPTIVE ROBOTIC HAND BASED ON IDLE-STROKE MECHANISM

Feb. 2026

H. Ding, H. Luan, T. Yang*, W. Zhang*

IEEE ICCMA 2025, Macau, China

[P3] GLINT: AN IDLE-STROKE GRASP-AND-LIFT HAND FOR IN-HAND MANIPULATION

Feb. 2026

H. Ding†, J. Fan, Z. Zhu, Y. Zhao, K. Chen*, W. Zhang*

IEEE ICARM 2025, Hangzhou, China

[P4] PEAUCELLIER GRIPPER: A NOVEL UNDERACTUATED GRIPPER FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASP

Dec. 2025

H. Ding†, Y. Chen†, D. Jia, W. Zhang*

IEEE ICIA 2025, Wuhan, China

[P5] AN UNDERACTUATED GRIPPER WITH GRASSHOPPER-INSPIRED LINKAGES AND DELAY TRIGGERING FOR PINCHING AND ADAPTIVE GRASPING

Dec. 2025

H. Ding†, W. Zhang*

IEEE/RSJ IROS 2025, Hangzhou, China

[P6] A NOVEL GRIPPER WITH SEMI-PEAUCELLIER LINKAGE AND IDLE-STROKE MECHANISM FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

Nov. 2025

H. Ding[†], Z. Chen, T. Yang^{*}, W. Zhang^{*}

[P7] SEMI-PEAUCELLIER LINKAGE AND DIFFERENTIAL MECHANISM FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

IEEE CASE 2025, Los Angeles, USA
Sep. 2025

OTHER PUBLICATIONS

Y. Liu[†], H. Ding[†], H. Chen, H. Gao, J. Yu, F. Mo, N. Wang^{*}

[P8] RECENT PROGRESS ON THE RESEARCH OF 3D PRINTING IN AQUEOUS ZINC-ION BATTERIES

Polymers (MDPI) 17(15):2136
Aug. 2025

H. Xie, L. You, X. Fang, L. Li, H. Ding, Z. Zhou, G. Zhang, D. Zhang

[P9] RAPID ANALYSIS OF HEAVY-METAL ELEMENT ADSORPTION BY SCG BASED ON LIBS TECHNOLOGY

E3S Web of Conferences, 615 (2025) 02010
Feb. 2025

Honors & Awards

INTERNATIONAL AWARDS

2025

Undergraduate Champion, ASME Student Mechanism & Robot Design Competition

USA

NATIONAL & DOMESTIC AWARDS

2025

National Scholarship, Ministry of Education of China

China

2024

President's Award, Shenzhen Technology University

Shenzhen, China

2024

Research and Innovation Award (First Prize), Shenzhen Technology University

Shenzhen, China

2023

Gold Prize, 9th China "Internet+" Innovation Competition (Guangdong Division)

Guangdong, China

2023

Second Prize, 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division)

Guangdong, China

2023

Research and Innovation Award (First Prize), Shenzhen Technology University

Shenzhen, China

Professional Service

2025

Conference Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2025

Session Chair, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Hangzhou, China

2025

Conference Reviewer, IEEE International Conference on Automation Science and Engineering (CASE)